

Introduction to Agent-Based Simulations

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KTH, Flemingsberg

Today's objectives

- What are Agent-Based Simulations?
- Where are they used?
- Why do they work?
- Relation to DES and SD
- How can it be used on MedTech

But first

- Non-linear systems / Complex Systems
- Emergence

Finding population growth Logistic Map

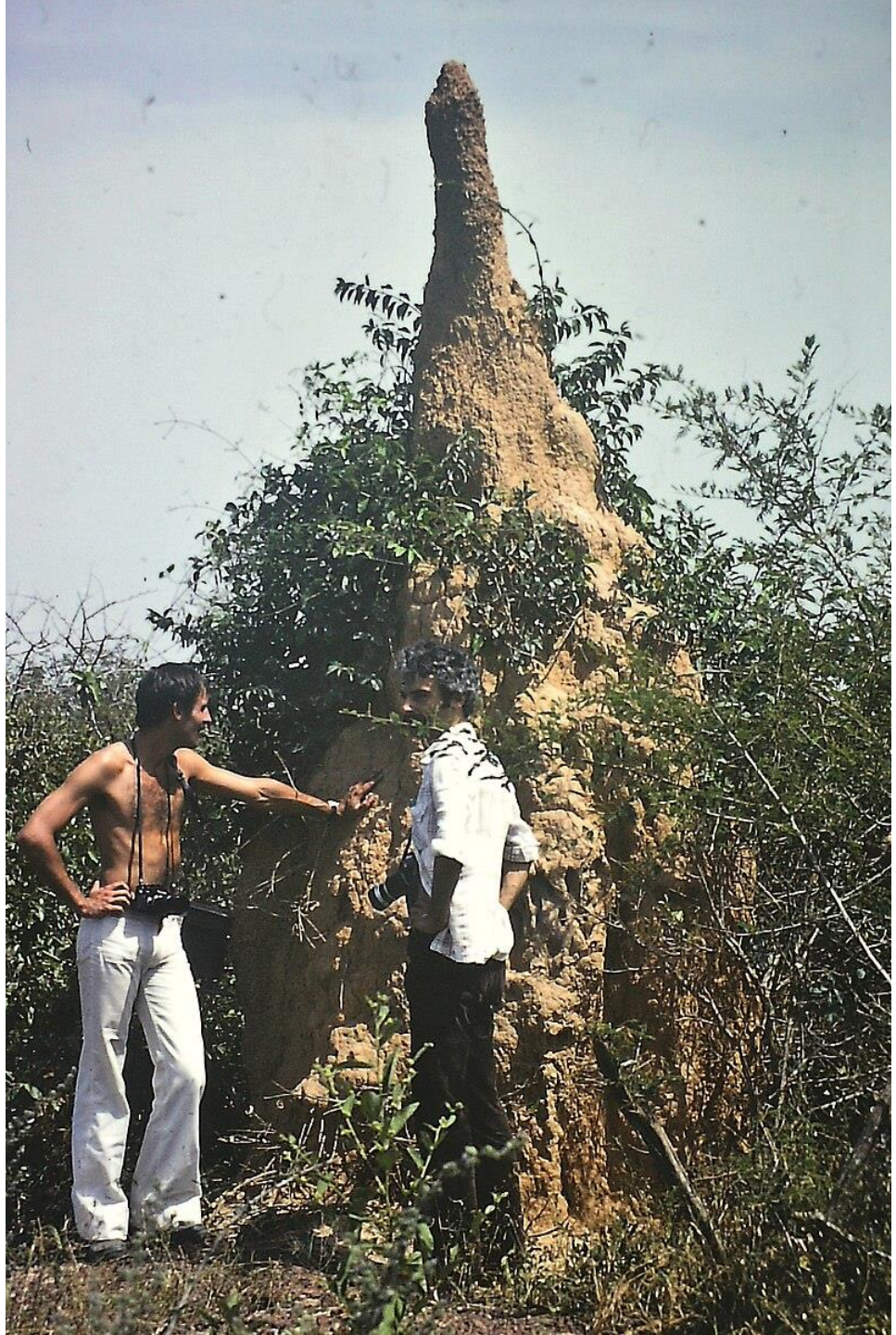
- Birth rate and death rate to find a progression of populations
- $x_{n+1} = r x_n (1 - x_n)$
- What happens at r values when 3 > x < 4 ?
- *Python Script*

Emergence

- NETLOGO
- Flocking

Termite mounds

- Soucre: Wikipedia
- Emergence: *When the sum of parts is more than its components.*
- Strong vs weak Emergence.
  - Weak: Independent identity maintained. Tractable and hence can be simulated
  - Strong: Identity lost, a new entity.



Complex systems properties

- Non-linearity
- Feedback

Why does it affect us?

- Traffic Jams, Braess's paradox, Sometimes the longer route is faster! (Cheonggye Expressway)
- Cobra Effect, Economic Economics (The whole field of behavioral economics!)
- Distributing Vaccines
- Spread of diseases
- Cellular Phones
- Urban growth
- Games

Agent-Based Models

- ”Agent” is an abstract notion that describes an entity with agency, i.e., its independence to perform actions.
- ABMs model agents' behavior over time or a series of steps, and we see the effect from a systemic perspective.
- Components of an ABM
  - Agents, of course! Populations of agents.
  - A universe for the agents to ”live in.”

Building a agent

- Two main frameworks
- BDI: Belief, Desire, and Intention approach.
  - Logic/Classical AI-based approach.
  - Uses logical equations to model
- Strategic Agents
  - Uses a game-theoretical framework.
  - ´Strategy Set, Utilities (function), Payoffs
  - Assume agents are rational within some bounds. Bounded Rationality (won't jump off a cliff!)
  - Each agent tries to maximize their ”utility.”
  - Examples of strategic games:
    - Prisoners Dilemma, Battle of the sexes
    - Auctions
    - Mechanism Design
- Modeling the universe
  - Wrap-around space
  - GIS-space
  - Discrete space and Cellular Automata
  - No space

Cellular Automata

- One case of ABMs.
- Discrete patches that depend on the state of neighbors' states.
- Conway's game of life
  - Scientific American, Vol. 223, No. 4 (October 1970), pp. 120-123
  - To play: https://conwaylife.com
  - Any live cell with fewer than two live neighbors dies, as if by underpopulation.
  - Any live cell with two or three live neighbors lives on to the next generation.
  - Any live cell with more than three live neighbors dies, as if by overpopulation.
  - Any dead cell with exactly three live neighbors becomes a live cell, as if by reproduction.

Salient features of ABMs

- Meso-scale simulations that connect individuals to the larger systems.
- Generative approach to simulations. (vs the reductive approach).
- Explain processes vs Predictions.

Disadvantages

- Very sensitive to starting conditions
- Needs a lot of data and lots of compute power.
- Validating ABM simulations can be hard.
- Lower predictive power

They are the same thing!

- Agent-based simulations, Discrete Event Simulations, Systems Dynamics Models, and Serious Games can be interchanged.
- Each is used in different contexts.
- ABMs are used for explanations and the effect of many different things.
- DES for logistics, or there is a definite order of events
- SD in case of continuous effect (analog).

- Based on the audience and the simulation scope, one of them is chosen.
- Based on which part of the system needs to be explored, AMBs are mesoscale.

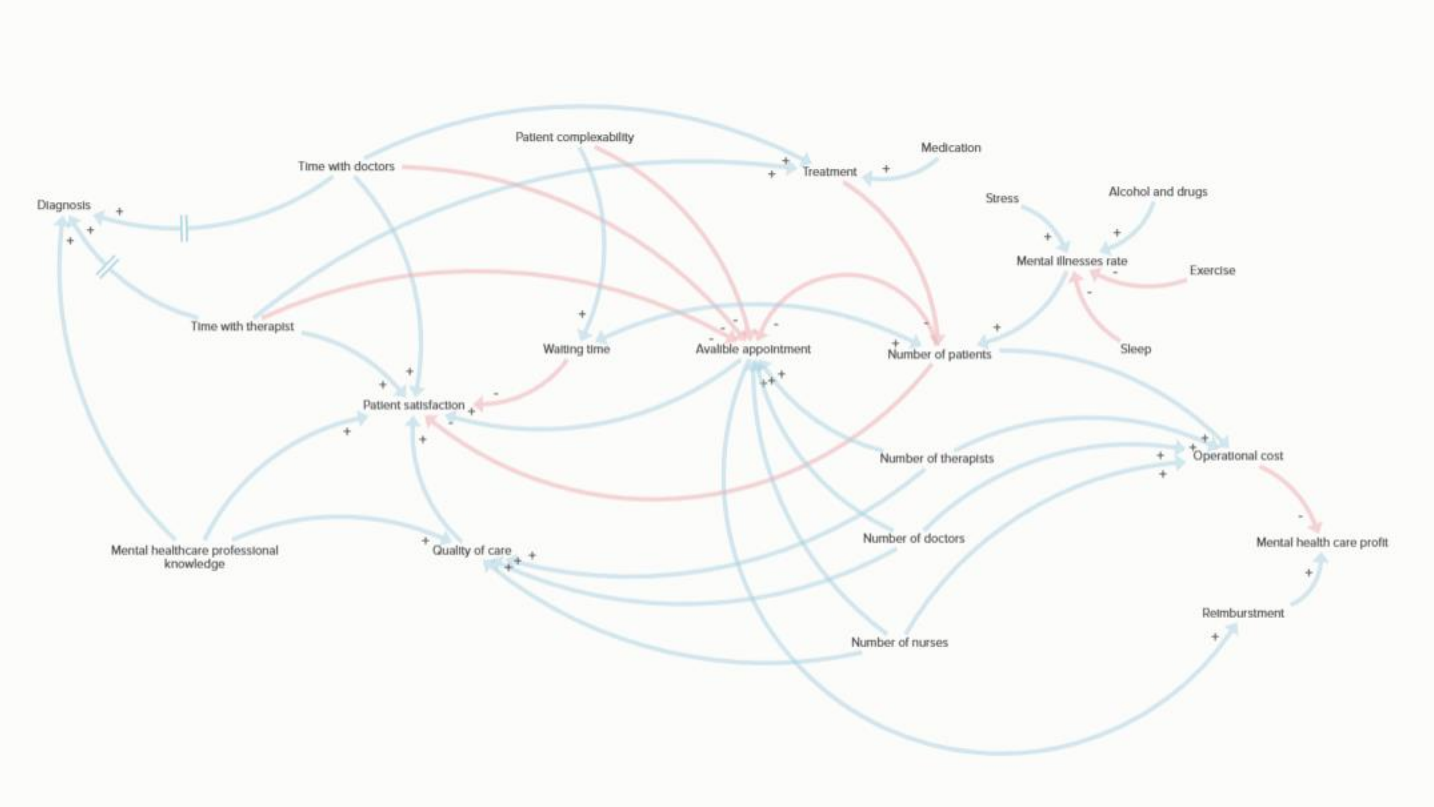
Where can they be used

- Logistics.
- Policy simulations
- Geographical Simulations with discrete components.
- Look for conditions where agent behavior depends on its neighbors, and there are many types of agents. OR there are human beings involved to be simulated.

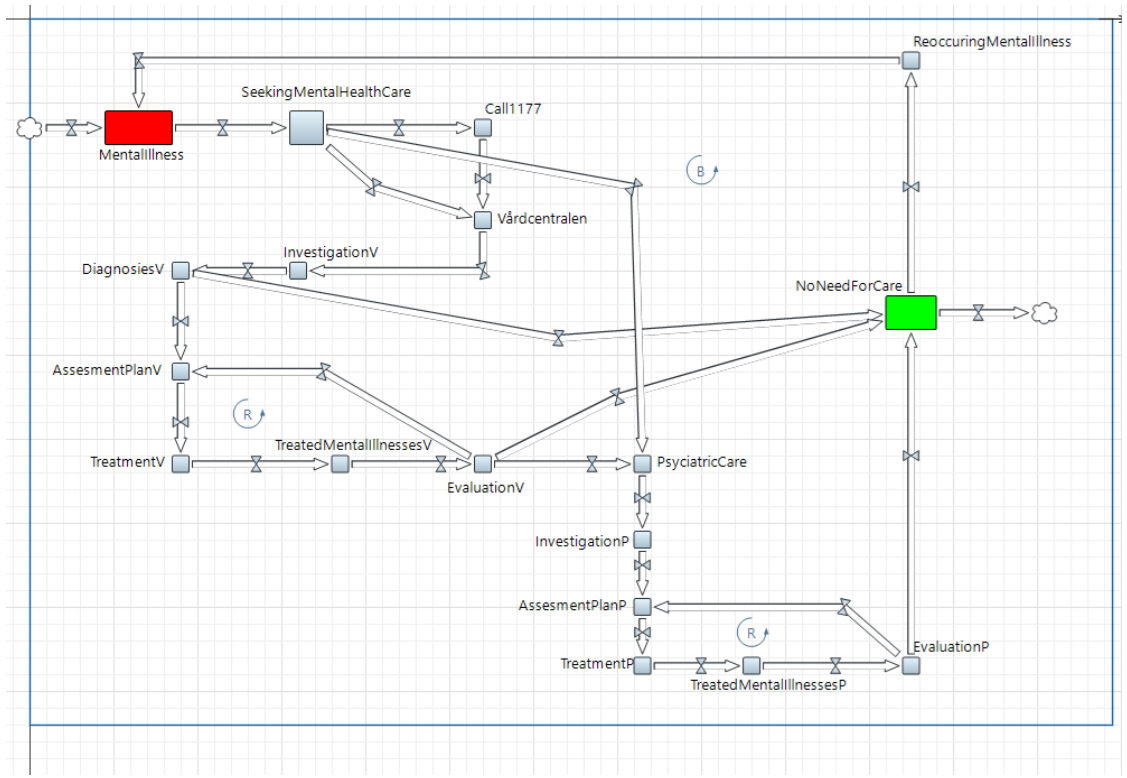
Some interesting books

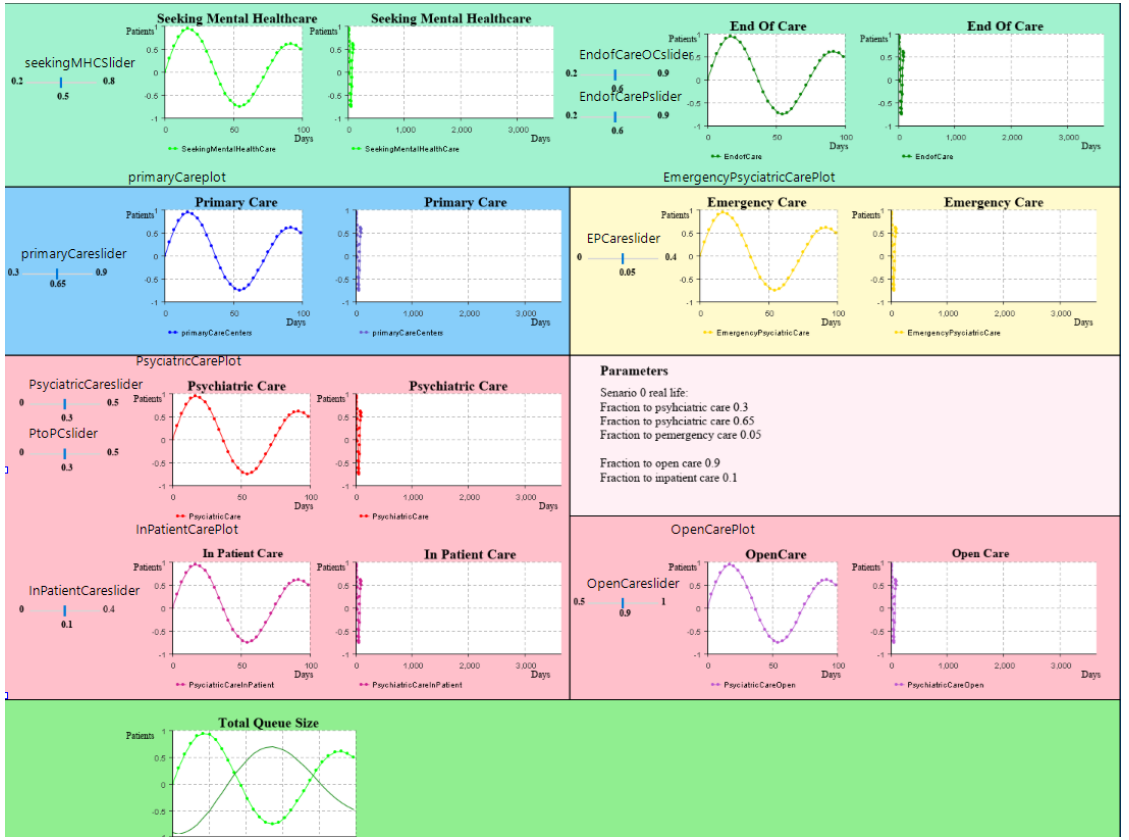
- Micromotives and Macro Behaviour, Thomas Schelling
- A new kind of Science, Stephen Wolform
- Cities And Complexity: Understanding Cities With Cellular Automata, Agent-based Models, And Fractals, Michael Batty
- Strategy: An Introduction to Game Theory, Joel Watson.
- Generative Social Science: Studies in Agent-Based Computational Modeling, Joshua M. Epstine

Model of Mental Healthcare Structure in Stockholm



Model of Mental Healthcare Structure in Stockholm





[DiVA, id: diva2:1837095]

Osswald J. Development of a System Dynamic Model of Mental Healthcare Structure in Stockholm [Student thesis]. TRITA-CBH-GRU. 2024.]


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